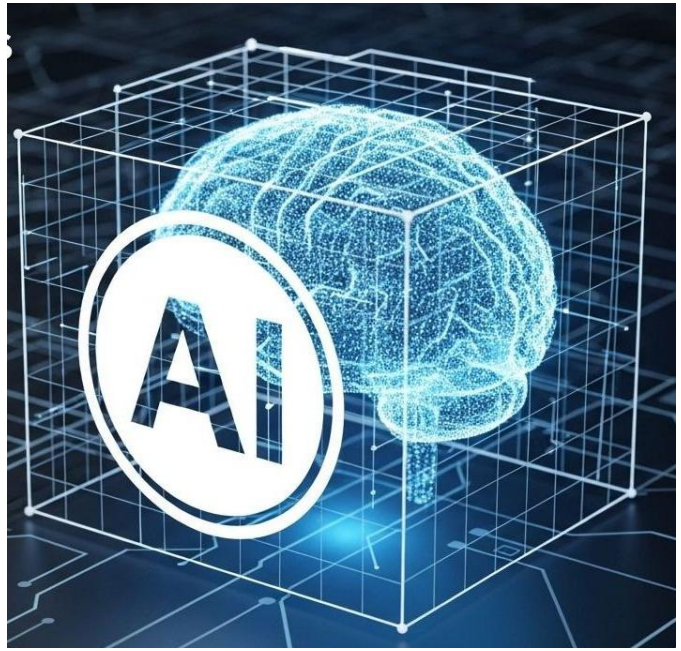


ALY 3510 · Fall 2026

Lecture 8: Robotics & Embodied AI

Intelligence that acts in the physical world

Pre-Lecture Video (4:54)



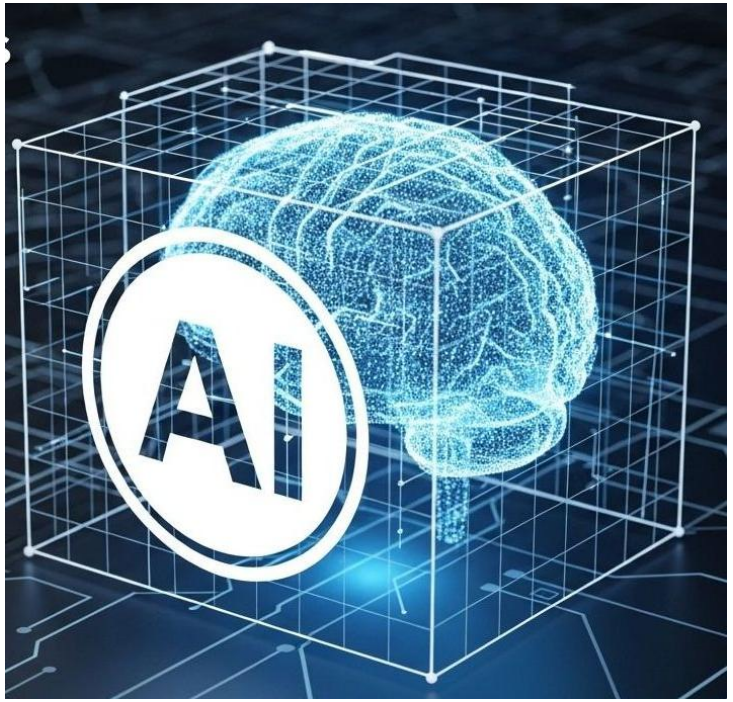
https://www.youtube.com/watch?v=8_IfxPI5ObM

Agenda

Pre-lecture video

Lecture

In-class discussion



Part 1: Introduction

What makes a robot intelligent?

What is a Robot? The Sense-Think-Act Cycle

A robot is a physical machine (agent) that senses its environment, decides what to do, and acts upon the world.

SENSE
Gather data — cameras, LiDAR, touch

THINK
Make decisions — AI, planning

ACT
Execute actions — motors, grippers, wheels

↺ **This cycle repeats continuously.**

Example: a Roomba senses obstacles, plans a path, and moves to clean the floor. Other examples: assembly arms, surgical robots, self-driving cars.

Evolution & Embodied AI

1960s
Shakey
First mobile reasoning robot

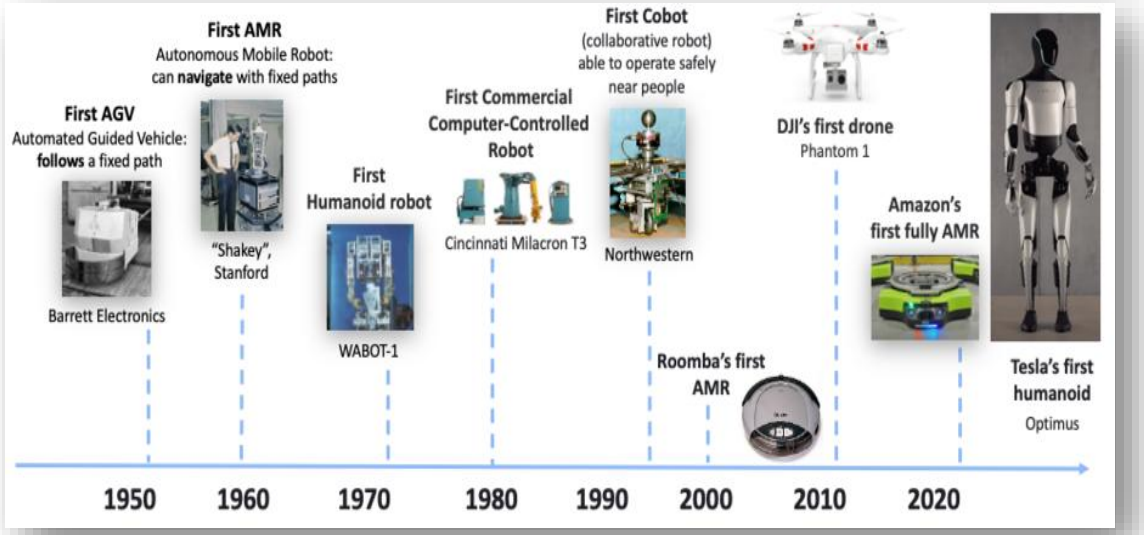
1980s
Industrial
Factory automation

2000s
Consumer
Roomba enters homes

2010s
Deep Learning
Vision breakthrough

2020s
LLMs
Natural language

Key Insight: Robots evolved from executing fixed programs to learning systems that improve through experience.



https://www.calcalistech.com/ctechnews/article/bycmlhvha#google_vignette

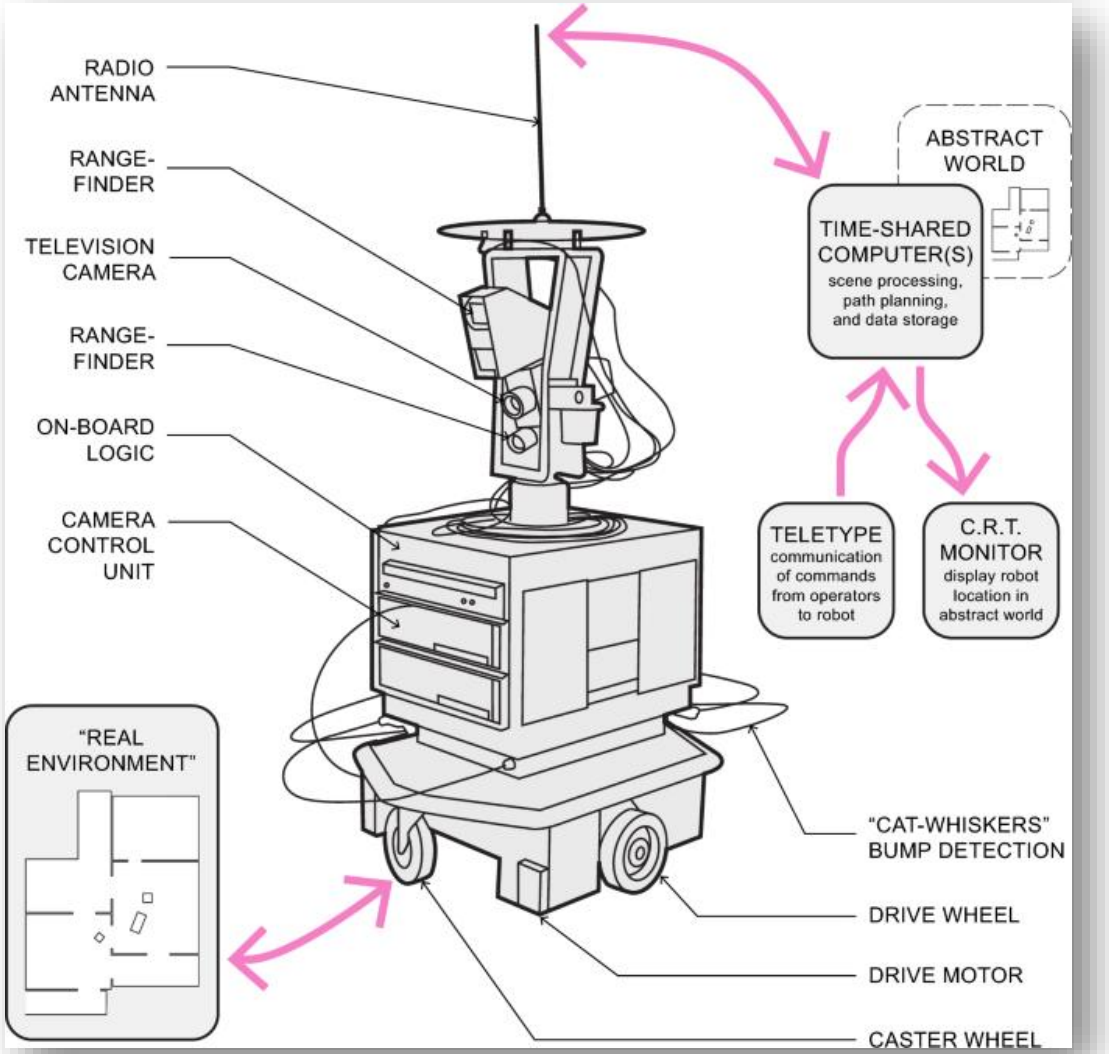
Embodied AI

- Intelligence integrated into a physical body that learns through real-world interaction.
- Must handle gravity, friction, uncertainty, noise, and real-time complexity.
- **Harder than software AI — often called "AI-complete."**

Robotic Intelligence: The Pioneer

1960s: Stanford Research Institute developed Shakey. This boxy robot could perceive its environment, model it, plan behavior, and execute tasks – a revolutionary achievement for its time.





https://link.springer.com/article/10.1007/s41693-024-00129-7

Part 2: Robot Hardware

Sensing and acting in the physical world

Robot Sensors: How Robots Perceive

Passive sensors

- Detect existing energy
- Cameras (light), microphones (sound)
- Temperature, touch/pressure

Active sensors

- Emit energy, measure reflection
- LiDAR (laser), ultrasonic (sound)
- Radar, infrared proximity

Sensors by function:

Vision

RGB, depth, IR cameras

Distance

LiDAR, ultrasonic, radar

Touch & force

Pressure, tactile, force

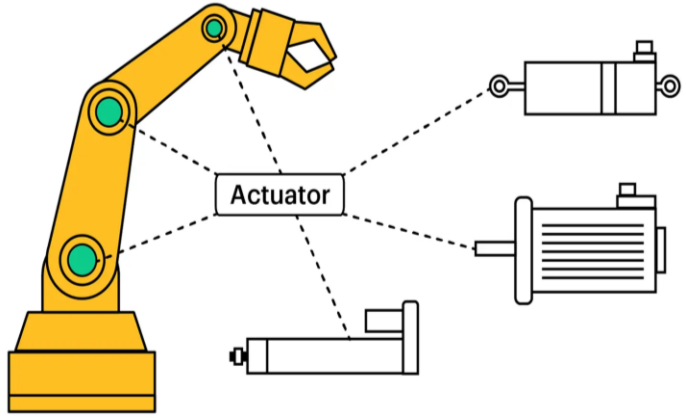
Motion & position

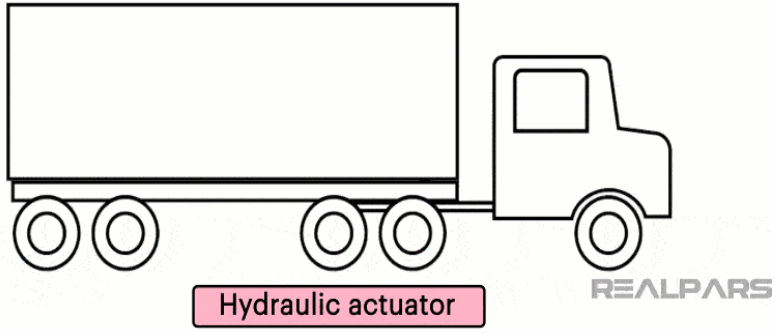
Gyros, accelerometers, GPS

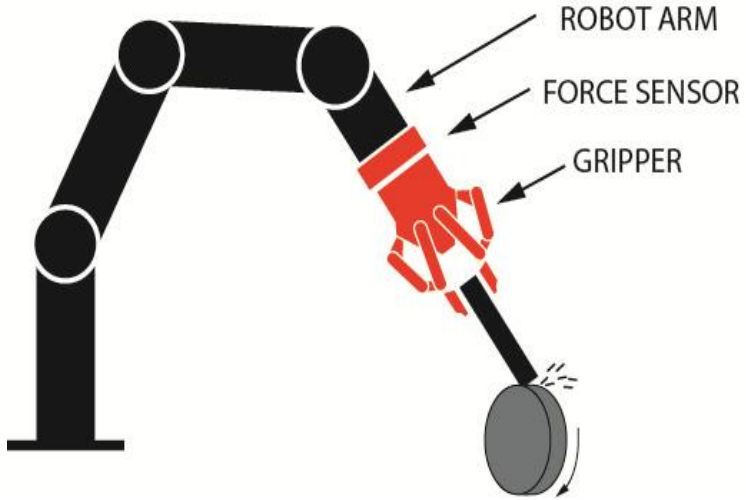
Sensor fusion — combining multiple sensor types — improves accuracy and reliability.

Actuators & Effectors: Robot "Muscles"

Type	Strengths	Used in
Electric motors	Precise, clean, easy to integrate, low power	Most robots — arms, 3D printers, drones
Hydraulic	Very powerful, high force	Heavy lifting, construction robots
Pneumatic	Fast, lightweight, safe around humans	Soft robots, grippers, medical devices







Emerging: shape-memory alloys, pneumatic artificial muscles, and piezoelectric actuators for ultra-precise motion.

Robot Types, Soft Robotics & Learning

Industrial
Factory automation (FANUC, ABB, KUKA)

Service
Assist humans — vacuums, delivery

Humanoid
Human-like form (Atlas, Optimus, Phoenix)

Mobile
Navigate environments — drones, rovers, AMRs

Soft robotics
Flexible, compliant, bio-inspired — safe, delicate handling

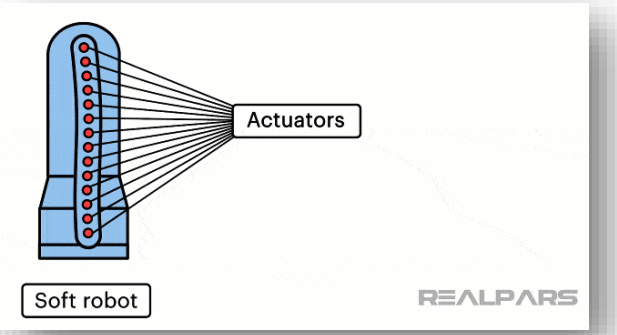
Sim-to-real
Train in simulation → bridge the gap → deploy in the real world











Robot Learning: Simulation to Reality

Training robots safely and efficiently in virtual environments

Simulation

Train in virtual world

→

Sim-to-Real

Bridge the gap

→

Real World

Deploy & refine

Why Simulate First?

- Safe: No risk of damage
- Fast: Run millions of trials
- Cheap: No physical wear
- Scalable: Train in parallel



Part 3: Digital Twins

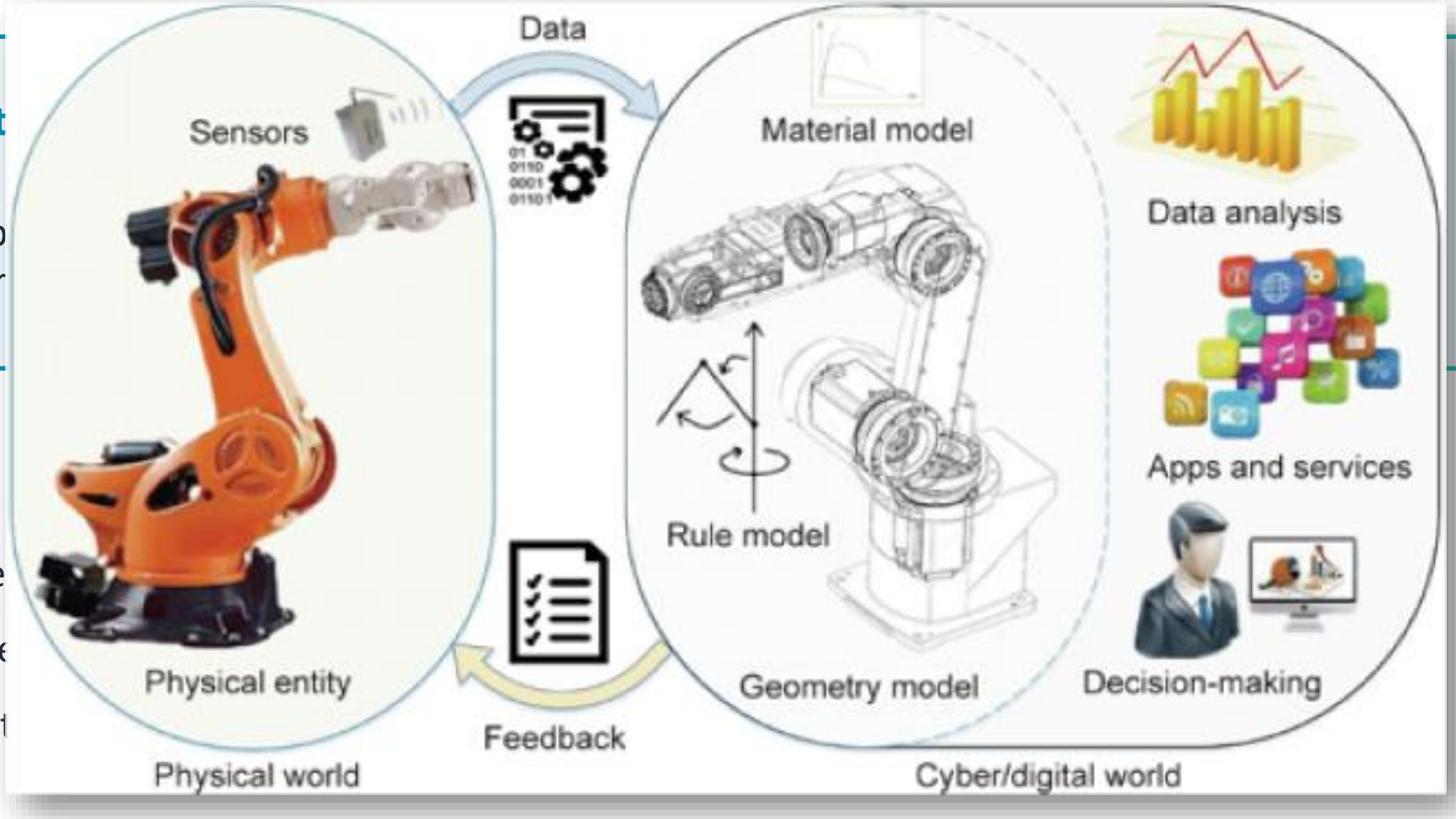
Virtual replicas of physical systems

What is a Digital Twin?

A virtual replica of a physical object, process, or system that is continuously updated with real-time data

Physical Syst

Real robot, factory, or p
physical wor



Key Capabilities

- Monitor system health re
- Predict failures before the
- Test changes virtually first

Digital Twins: Cyber-Physical Systems

The twin closes the loop: live sensor data in, simulation and prediction, decisions back out.

Domain	Digital-twin uses
Manufacturing	Factory-floor simulation, production & quality optimization, predictive maintenance
Robotics	Behavior testing, path-planning validation, collision detection, remote monitoring
Infrastructure	Smart-city planning, traffic-flow modeling, building & energy-grid management

Gartner (2017) forecast that half of large industrial companies would use digital twins by 2021, gaining ~10% in effectiveness.

Source: Gartner, "Digital Twins Will Impact Economic and Business Models," 2017.

Digital Twin: How it Works (1:56)



What to Watch For:

- How physical and digital systems stay synchronized
- Real-time data flow between twin and physical asset
- Predictive capabilities enabled by the digital model

https://www.youtube.com/watch?v=iVS-AuSjpOQ&t=2s

BREAK: "Do You Love Me?" (2:54)



https://www.youtube.com/watch?v=fn3KWM1kuAw

Part 4: Robot Motion & Autonomy

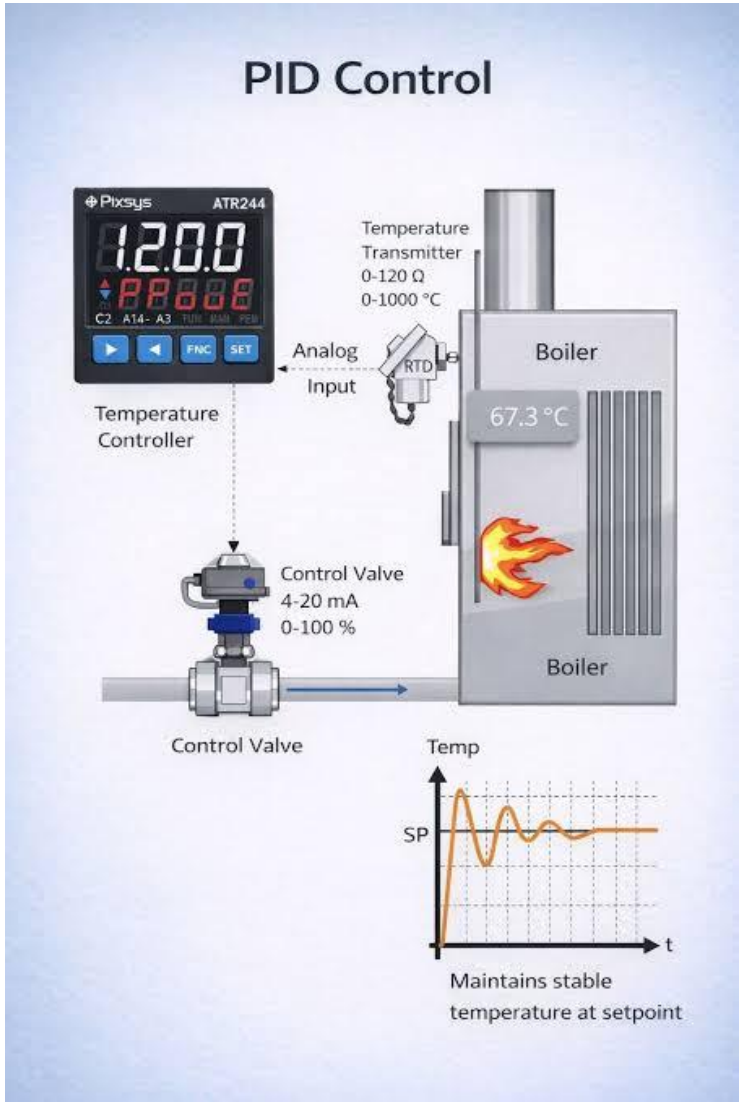
Control, planning, and navigation

Control Hierarchy & PID

Level	Role	Speed
High-level	Mission planning, task & goal setting	Slow, abstract
Mid-level	Path planning, obstacle avoidance	—
Low-level	Motor commands, real-time feedback (PID)	Fast, concrete

PID — the workhorse of low-level control

- P (proportional): how far am I from the target?
- I (integral): how long have I been off target?
- D (derivative): how fast am I approaching?

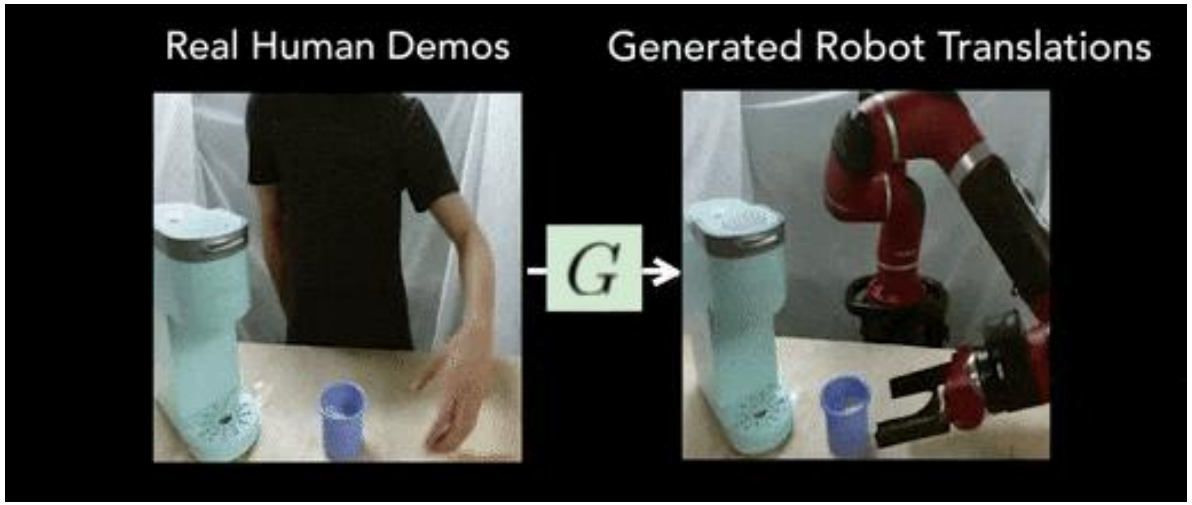


https://share.google/1icACUupp6Ff7QxOk

From Traditional to AI-Powered Control

Traditional control

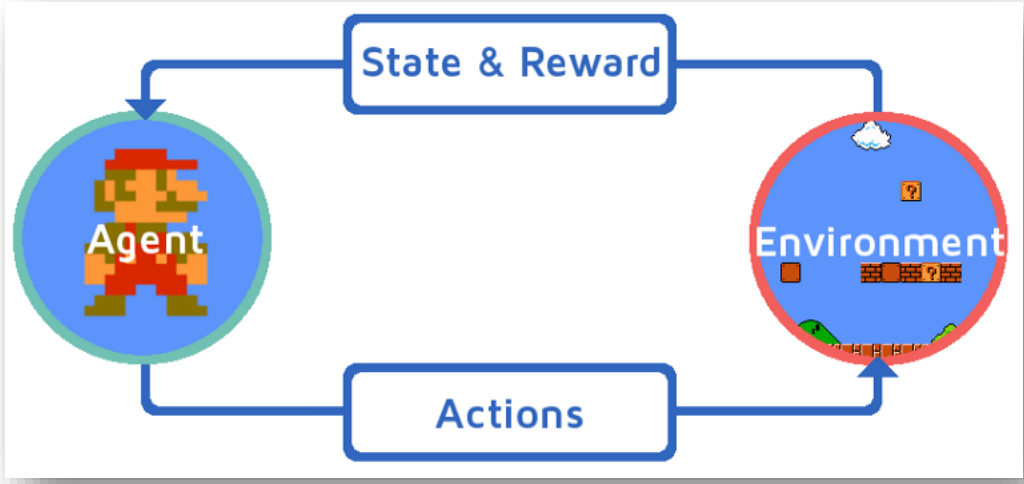
- Hand-tuned rules & feedback loops
- Predictable; needs manual tuning
- Great for well-defined tasks



https://bair.berkeley.edu/blog/2019/12/13/humans-cyclegan/

AI-powered control

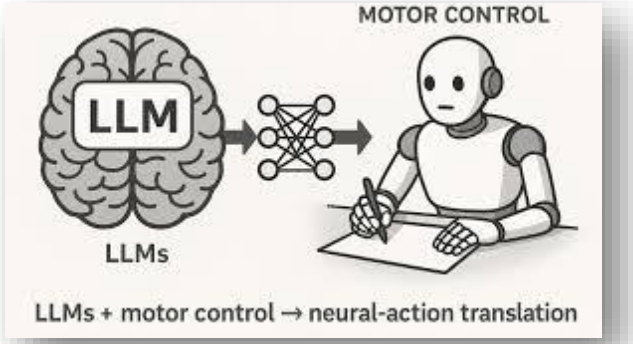
- Learns adaptive behavior from experience
- Reinforcement / imitation learning
- Handles complexity — but is data-hungry



https://www.freecodecamp.org/news/a-brief-introduction-to-reinforcement-learning-7799af5840db/

LLMs as the robot's "brain": translate human intent into robot-executable plans — natural-language commands, task breakdowns, and error recovery (e.g., Google DeepMind Gemini Robotics 2, NVIDIA GR00T N1.7, Figure Helix 02).

How LLMs Enhance Robots



Large Language Models bring natural interaction and reasoning to robotics

Natural Language Commands

"Pick up the red cup and put it next to the plate"

Task Planning & Reasoning

Break complex goals into executable robot actions

Context Understanding

Interpret ambiguous requests using world knowledge

Error Recovery

Explain failures and suggest alternative approaches

SLAM & Autonomous Navigation

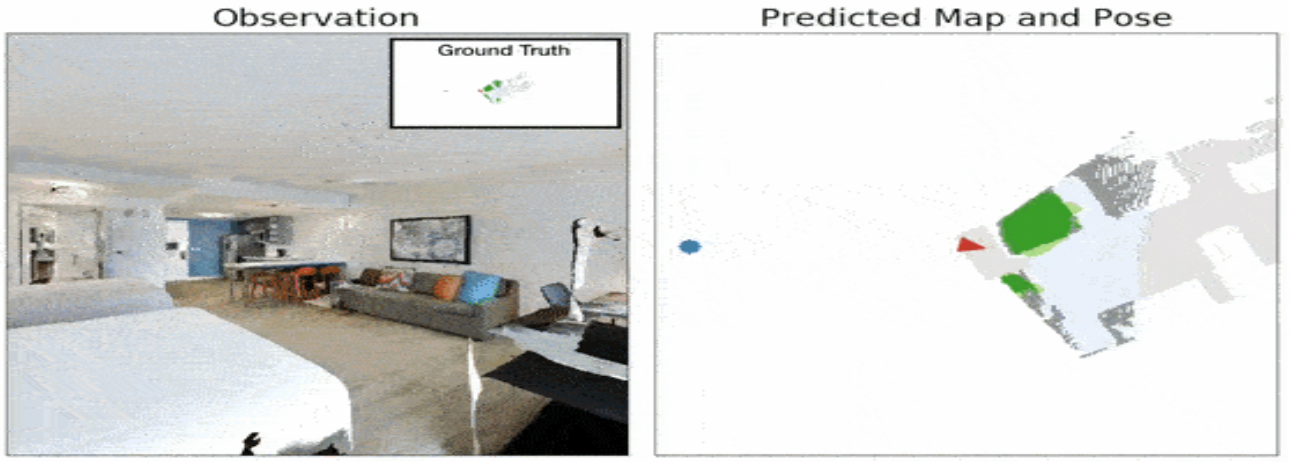
SLAM — Simultaneous Localization & Mapping: build a map of an unknown environment while tracking your own location in it.

The chicken-and-egg problem

- To build a map, you need to know where you are
- To know where you are, you need a map
- SLAM solves both at once (probabilistic methods)

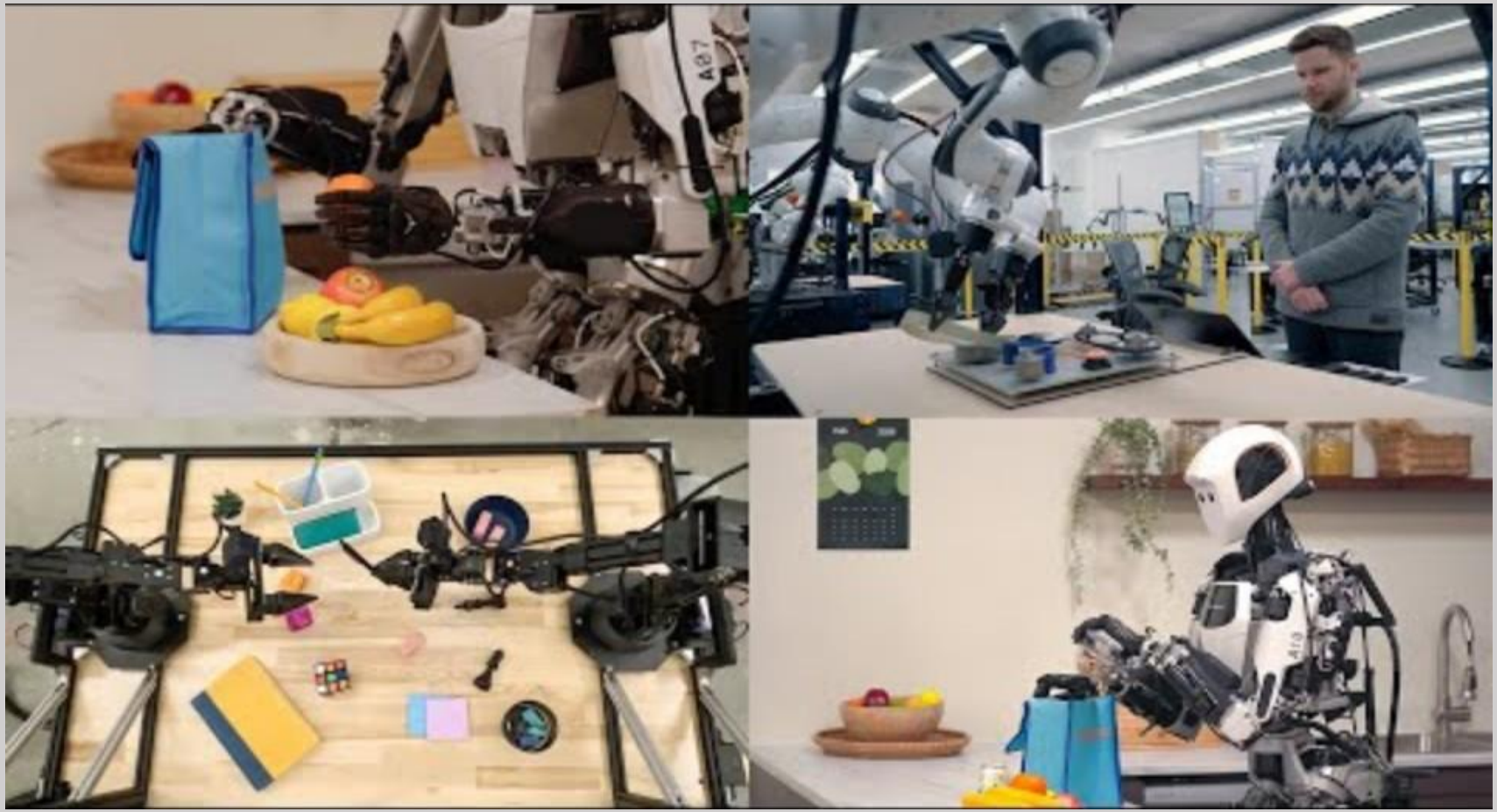
Navigation pipeline

- Perception → localization → planning → control
- Path planning: A*, RRT, potential fields
- Obstacle avoidance: reactive + velocity limits



SLAM is fundamental to anything that navigates unmapped space — vacuum robots, warehouse AMRs, drones, AR/VR headsets, Mars rovers.

Gemini Robotics: Bringing AI to The Physical World (2:59)



What to Watch For:

- How robots perceive and interact with objects
- Natural language understanding for task instructions
- Adaptability to different environments

https://www.youtube.com/watch?v=4MvGnmmP3c0&t=24s

Part 5: Autonomous Vehicles

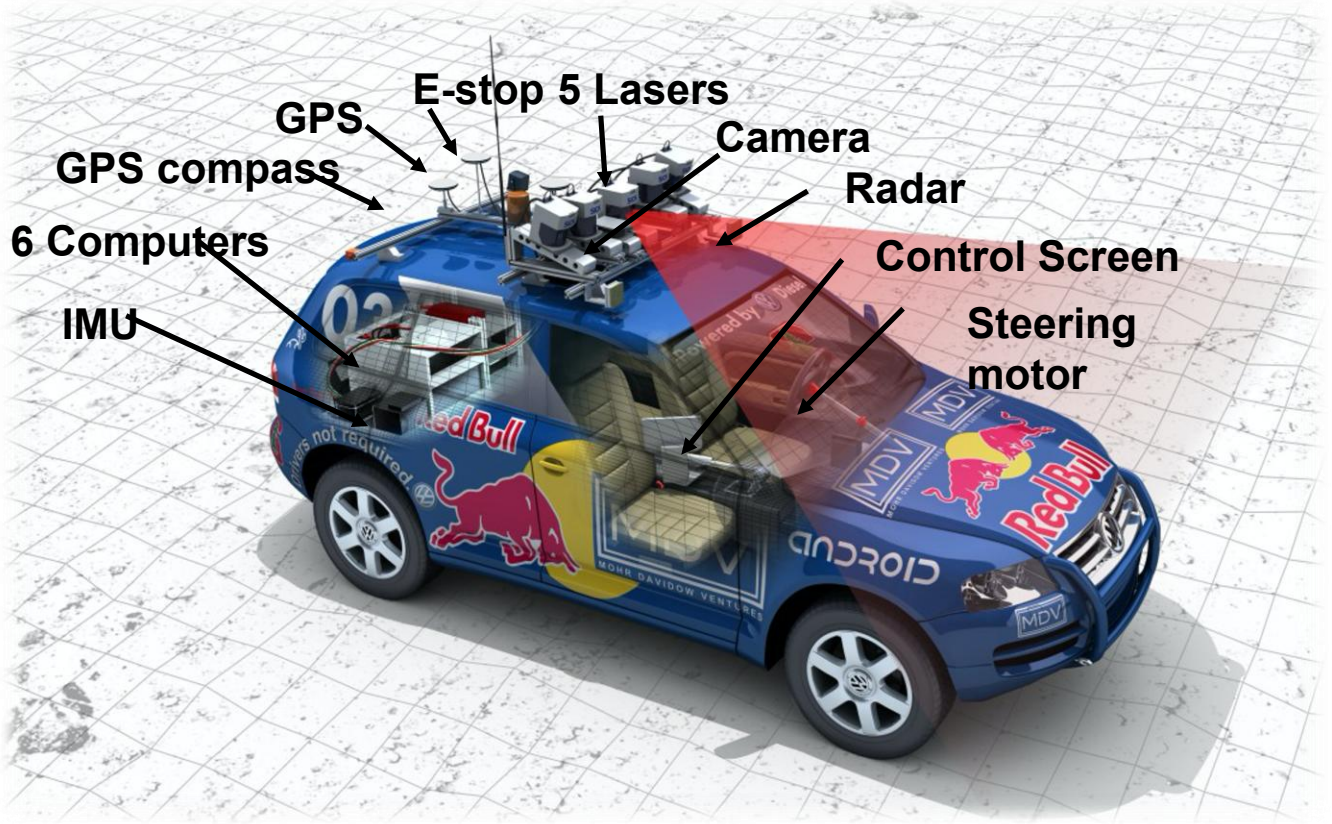
Self-driving technology & challenges

Self-Driving: Sensors & Perception

Multi-sensor fusion for robust perception:

- Cameras — color, signs, lane markings
- LiDAR — laser pulses build a 3D point cloud (cm-level)
- Radar — works in rain/fog; ultrasonic — close range

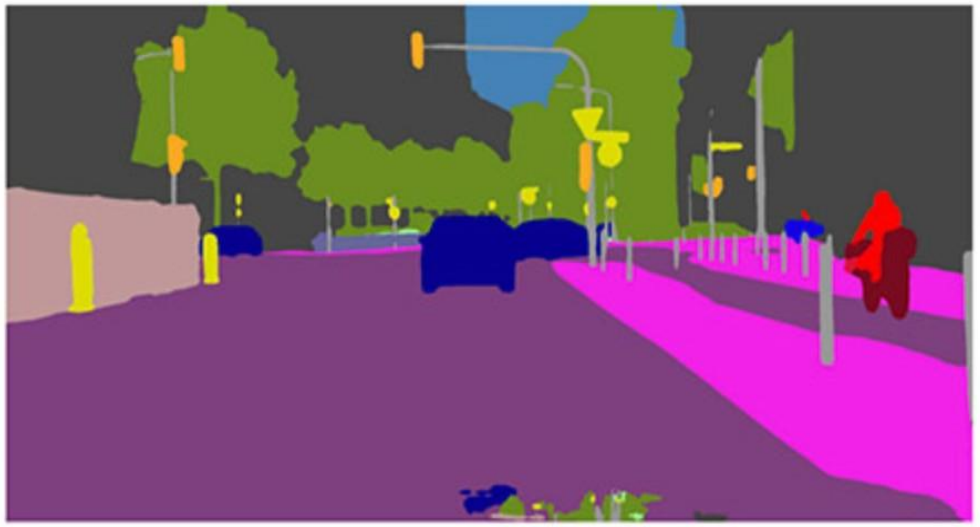
Perception (detect, segment, track) reuses the computer-vision methods from Lecture 7.



Semantic Scene Segmentation

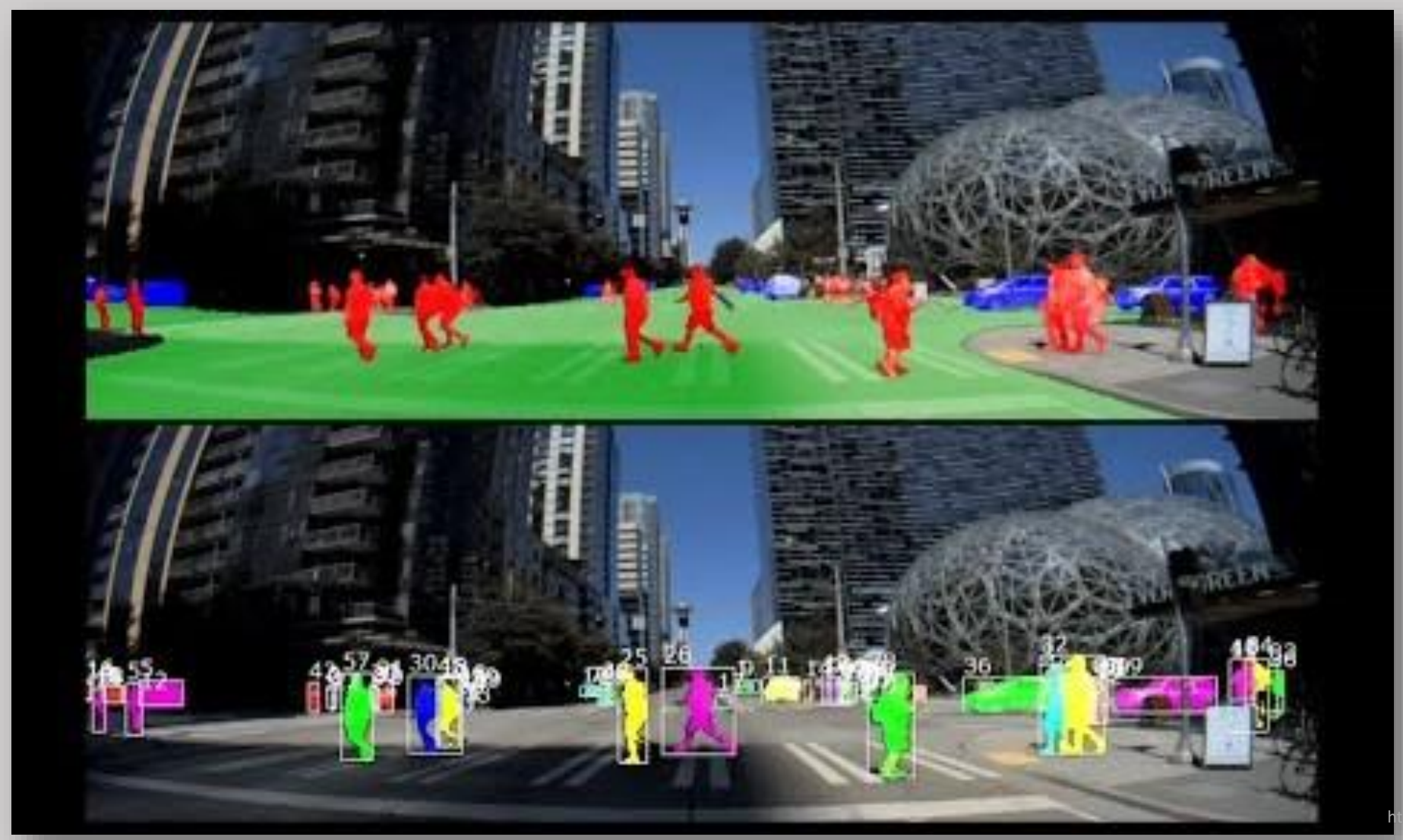
Every pixel is classified – Vision AI in action



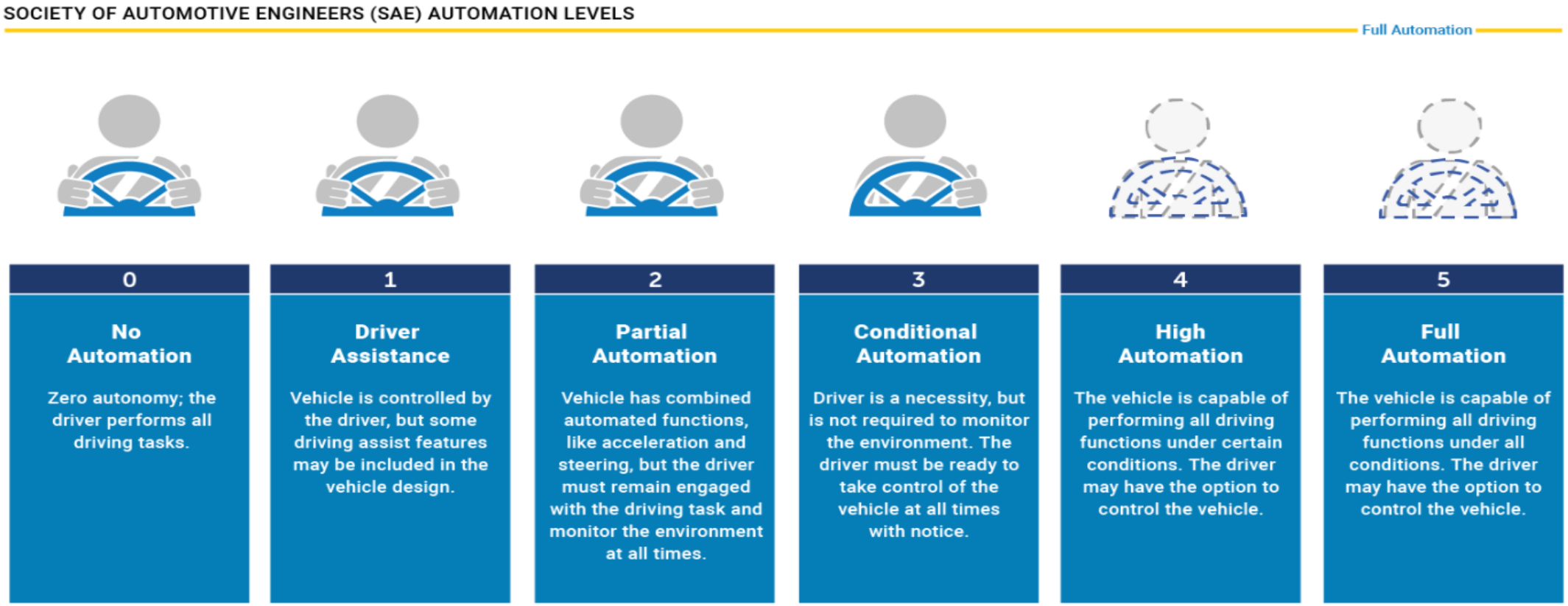
Panoptic Segmentation (1:27)

"Masks" are created. Object instances are detected and tracked.



https://www.youtube.com/watch?v=HS1wV9NMLr8&t=4s

SAE Levels & Urban Challenges



SAE J3016 — Levels 0–5. Level 3+ shifts responsibility from human to system

The "long-tail" problem:

Rare events are infinite in variety — unpredictable pedestrians, complex intersections, weather, faded lane markings, regional norms. Each new edge case needs new training data or explicit handling.

Part 6: Robot Manipulation

Grasping, dexterity, and physical interaction

Grasping, Degrees of Freedom & Hands

Humans perform roughly 4,000–7,000 grasps a day without thinking about it[1].
Robots still struggle with novel objects — grasping is surprisingly hard.

Why it's hard

- Object recognition & pose estimation
- Grasp-point selection; force control
- Handling transparency, reflections, clutter

Hands & degrees of freedom

- Simple gripper (1 DoF) → human hand (27+ DoF)
- End effectors: parallel-jaw, suction, magnetic
- Dexterous hands: Shadow, Allegro, Tesla

Modern approaches: deep learning for grasp prediction, sim-to-real transfer, tactile sensing, compliant/soft grippers.

[1] *Bullock et al., "Grasp Frequency and Usage in Daily Household and Machine Shop Tasks," IEEE Transactions on Haptics, 2013.*

Dexterous Manipulation in Action (3:40)



What to Watch For:

- In-hand object rotation and repositioning
- Handling of various object shapes
- Recovery from failed grasps
- Finger coordination and timing
- Force control (gentle vs. firm)
- Speed and precision of movements

https://www.youtube.com/watch?v=xXbqqeUU7js

Part 7: Humanoid Robots

The quest for human-like robots

Why Humanoids? Key Players

Why build humanoids

- Fits human spaces: tools, stairs, doors
- Can learn from human demonstrations
- Natural to work alongside people



The hard part: walking

- Bipedal locomotion is “controlled falling”
- Balance, power draw, complex control
- Falls can cause damage

Key players, 2026[1]

- **Deployed:** Agility Digit, Figure 03 (BMW)
- **Piloting:** Boston Dynamics Atlas, Tesla Optimus, Sanctuary Phoenix and Apptronik Apollo
- **Research:** Unitree G1 / H1
- **Home:** 1X NEO (preorders)

1. Deployment status as of mid-2026. Source: company announcements.

https://www.linkedin.com/pulse/humanoid-cambrian-explosion-why-2026-year-industrial-m%C3%B3zes-uaabf/

Part 8: Human-Robot Interaction

Working safely alongside people

Human-Robot Interaction: Safety & Social Intelligence

Safety first

- Force/torque limiting; speed & separation monitoring
- Soft/compliant materials; emergency stops
- Cobots: designed to share space (ISO 10218/15066) with humans without safety cages

Social intelligence

- Intent recognition — predict what humans do
- Natural-language & emotion understanding
- Respect social norms; build trust through predictability



https://iliad.stanford.edu/research/humanrobot



https://iliad.stanford.edu/blog/2020/03/19/when-humans-arent-optimal-robots-that-collaborate-with-risk-aware-humans/

Current Challenges & What Lies Ahead

What's holding robotics back

- **Hardware:** battery life, cost, hands
- **AI:** coping with new tasks and spaces
- **Safety:** verifying learned behavior

Where it's heading

- General-purpose, multi-task robots
- Home and eldercare assistants
- Robots for space and hazardous work

Key insight: hardware is improving steadily; safe, adaptable AI is the harder problem.

Open ethical questions: lethal autonomous weapons and human oversight.

Summary

Key takeaways and discussion

Key Takeaways

Embodied AI is different Robots act in a noisy physical world where mistakes have real consequences.

Sense–Think–Act Sensors perceive, AI and control (e.g., PID) decide, actuators move — in a loop.

Simulate first Digital twins and sim-to-real let robots train and be tested safely in software.

Foundation models as the brain LLMs turn plain-language commands into robot actions.

Autonomy is arriving unevenly Cars, cobots and humanoids deploy; adaptable AI is the bottleneck.

Let's Discuss (5–10 min)

1. *"We know how to build robots — teaching them what to do is the hard problem." Why is acting in the physical world so much harder for AI than answering questions on a screen?*

2. *Should machines ever make life-or-death decisions without human oversight — in a self-driving car's split-second choice, or an autonomous weapon? Where do you draw the line, and who enforces it?*

Alternate: Humanoid robots are expensive and hard to build. When is a human-shaped robot actually necessary — versus a wheeled or specialized one?